

EE451 PHW1
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For the entire PHW, the test platform used was the USC CARC Discovery cluster. The number of runs performed for each problem was 1. In terms of other necessary variables, the compiler used was g++.

For p1a, the execution time and performance is shown in the output below:

```
[[rafaelwa@discovery1 PHW1]$ ./p1a  
Number of FLOPs = 17179869184, Execution time = 263.893 sec,  
65.1016 MFLOPs per sec  
C[100][100]=2.30093e+08
```

For p1b, the execution time and performance for b=8 is shown in the output below:

```
[[rafaelwa@discovery1 PHW1]$ ./p1b 8  
Number of FLOPs = 17179869184, Execution time = 149.999 sec,  
114.534 MFLOPs per sec  
C[100][100]=2.30093e+08
```

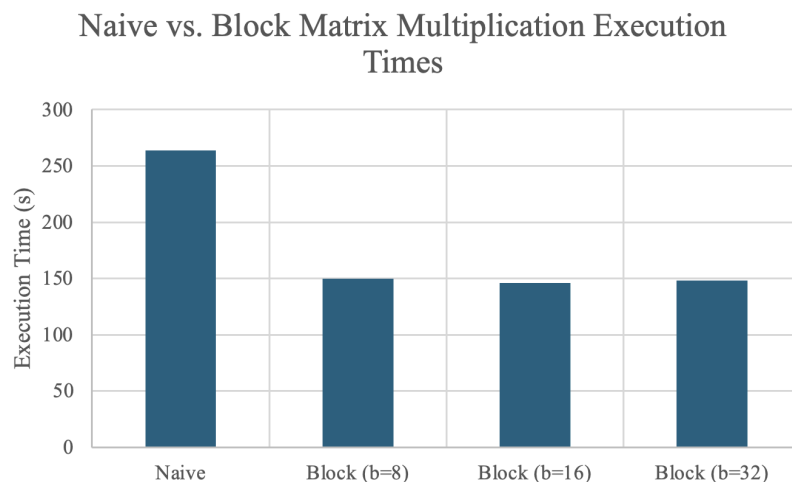
The execution time and performance for b=16 is shown in the output below:

```
[[rafaelwa@discovery1 PHW1]$ ./p1b 16  
Number of FLOPs = 17179869184, Execution time = 146.162 sec,  
117.54 MFLOPs per sec  
C[100][100]=2.30093e+08
```

The execution time and performance for b=32 is shown in the output below:

```
[[rafaelwa@discovery1 PHW1]$ ./p1b 32  
Number of FLOPs = 17179869184, Execution time = 148.105 sec,  
115.998 MFLOPs per sec  
C[100][100]=2.30093e+08
```

The plot comparing the result against the result using naive matrix multiplication is shown below:



It is observed that for all values of b , the block matrix multiplication significantly outperforms the naive matrix multiplication in terms of execution time. Increasing b from 8 to 16 provides marginal improvements in execution time. Increasing b from 16 to 32 results in an increase in execution time, which is likely a result of the cache size starting to be exceeded. I additionally experimented with smaller values, such as $b=2$ (execution time of 200.387 seconds) and $b=4$ (execution time of 171.156 seconds), illustrating how increasing the value of b improves execution time up to a certain point before it begins tapering off.

For p2, the execution time is shown in the output below:

```
[[rafaelwa@discovery1 PHW1]$ ./p2  
Execution time: 1.00773 seconds
```

The corresponding image is shown below:

